WORKSHEET 1.1

The Instant Problem: From Average Rates to Limits

Unit 1: Limits and Continuity

Key ideas

- The **average rate of change** of f on $[a, b]$ is $\dfrac{f(b) - f(a)}{b - a}$, with units of f per unit of x.
- An **instantaneous** rate is the limit of average rates as the interval shrinks to a point.
- $\lim\limits_{x \to c} f(x)$ describes the values of f *near* c. The value $f(c)$ plays no part.
- A two-sided limit exists exactly when the left-hand and right-hand limits exist and are equal.

Worked examples

Example 1 · Average rate from a formula

Find the average rate of change of $f(x) = x^2 - 5x$ on $[2, 6]$.

Solution. $f(6) = 6$ and $f(2) = -6$, so $\dfrac{6 - (-6)}{6 - 2} = \dfrac{12}{4}$.

The average rate of change is 3.

Example 2 · Shrinking the interval

For $s(t) = 2t^2 + t$, simplify the average rate of change on $[3, 3 + h]$, $h \neq 0$, and find what it approaches as $h \to 0$.

Solution. $\dfrac{2(3 + h)^2 + (3 + h) - 21}{h} = \dfrac{13h + 2h^2}{h} = 13 + 2h.$

As $h \to 0$ the average rates approach 13.

Example 3 · Interpreting in context

The depth of water in a pool, $D(t)$ cm, is measured at times t minutes.

t	0	4	10	15
$D(t)$	60	52	34	29

Find the average rate of change of D on $[4, 10]$ and interpret it.

Solution. $\dfrac{D(10) - D(4)}{10 - 4} = \dfrac{34 - 52}{6} = -3.$

From $t = 4$ to $t = 10$ minutes, the depth of the water decreased at an average rate of 3 cm per minute.

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Practice

CORE

1. Find the average rate of change of $f(x) = x^3$ on $[-1, 2]$.

2. Find the average rate of change of $f(x) = \dfrac{1}{x}$ on $[1, 4]$.

3. A function has $\lim\limits_{x \to 3^-} f(x) = 5$, $\lim\limits_{x \to 3^+} f(x) = 5$ and $f(3) = 1$. State $\lim\limits_{x \to 3} f(x)$.

4. A function has $\lim\limits_{x \to 0^-} g(x) = -2$ and $\lim\limits_{x \to 0^+} g(x) = 2$. Does $\lim\limits_{x \to 0} g(x)$ exist? Explain.

EXAM LEVEL

5. For $s(t) = t^2 - 3t$, simplify the average rate of change on $[2, 2+h]$, $h \neq 0$, and state the instantaneous rate of change at $t = 2$.

6. Let $g(x) = \dfrac{x^2 - 1}{x - 1}$ for $x \neq 1$ and $g(1) = 7$. Find $\lim\limits_{x \to 1} g(x)$.

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7. Using the pool table in Example 3, find the average rate of change of D on $[10, 15]$ and interpret it with units.

8. The average rate of change of f on $[2, 5]$ is -4, and $f(5) = 1$. Find $f(2)$.

Question 9 · Free-response practice

The volume of water in a tank, $V(t)$ litres, is recorded at selected times t minutes.

t	0	3	8	12
$V(t)$	200	170	130	122

(a) Find the average rate of change of V on $[3, 8]$. Indicate units of measure.

(b) Using correct units, interpret your answer to part (a) in context.

(c) On which of the intervals $[0, 3]$, $[3, 8]$, $[8, 12]$ is the water draining fastest on average? Show the work that leads to your answer.

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Worked solutions

1. $\dfrac{8-(-1)}{2-(-1)} = \dfrac{9}{3} = 3.$

2. $\dfrac{\frac{1}{4}-1}{4-1} = \dfrac{-\frac{3}{4}}{3} = -\dfrac{1}{4}.$

3. 5. The one-sided limits agree; the value $f(3) = 1$ is irrelevant.

4. No. The one-sided limits exist but are not equal.

5. $\dfrac{(2+h)^2 - 3(2+h) - (-2)}{h} = \dfrac{h + h^2}{h} = 1 + h$, which approaches 1.

6. For $x \neq 1$, $g(x) = x + 1$, so the limit is 2 (not $g(1) = 7$).

7. $\dfrac{29-34}{15-10} = -1$. From $t = 10$ to $t = 15$ minutes the depth decreased at an average rate of 1 cm per minute.

8. $\dfrac{1-f(2)}{3} = -4$ gives $f(2) = 13$.

9. (a) $\dfrac{V(8)-V(3)}{8-3} = \dfrac{130-170}{5} = -8$ litres per minute.

   (b) From $t = 3$ to $t = 8$ minutes, the volume of water in the tank decreased at an average rate of 8 litres per minute.

   (c) $[0,3]$: $\dfrac{170-200}{3} = -10$. $[3,8]$: -8. $[8,12]$: $\dfrac{122-130}{4} = -2$. The water drains fastest on $[0,3]$, at an average of 10 litres per minute.

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